GEOG 172

Global Spatial Statistics

Taking concepts and algorithms into the computer

Lecture 6-2

GLOBAL STATISTICS

These tests result in a single observed statistic that characterizes an observed map. We then compare that statistic with a critical value to decide whether the null hypothesis of spatial randomness should be rejected.

LOCAL STATISTICS

These tests are designed to test the null hypothesis of randomness around a prespecified point – typically against an alternative that there is a clustering of raised risk around the point.

SCAN STATISTICS

A window is centered on a particular location to see whether there is significant clustering at that location – and then the window is moved across the map to see if there is clustering around any location.

Moran's I

Understanding the mechanics of the method

MORAN'S I

The Moran's I statistics is a correlation coefficient, but unlike a traditional correlation coefficient that measures similarity or a relationship between two different attributes (e.g., income and education), the Moran's I statistic measures the relationship between the same attribute at different locations (e.g., income in location i and income at location j)

$$I = \frac{n \sum_i^n \sum_j^n w_{ij}(y_i - \bar{y})(y_j - \bar{y})}{\left(\sum_i^n \sum_j^n w_{ij}\right) \sum_i^n (y_i - \bar{y})^2}$$

UNCLE MORAN'S BLOCK PARTY

Where the party at?

ACTIVITY

Purpose: Calculate the mean, deviations, total squared deviations, neighboring deviation products, and Moran's I for a six-house arrangement.

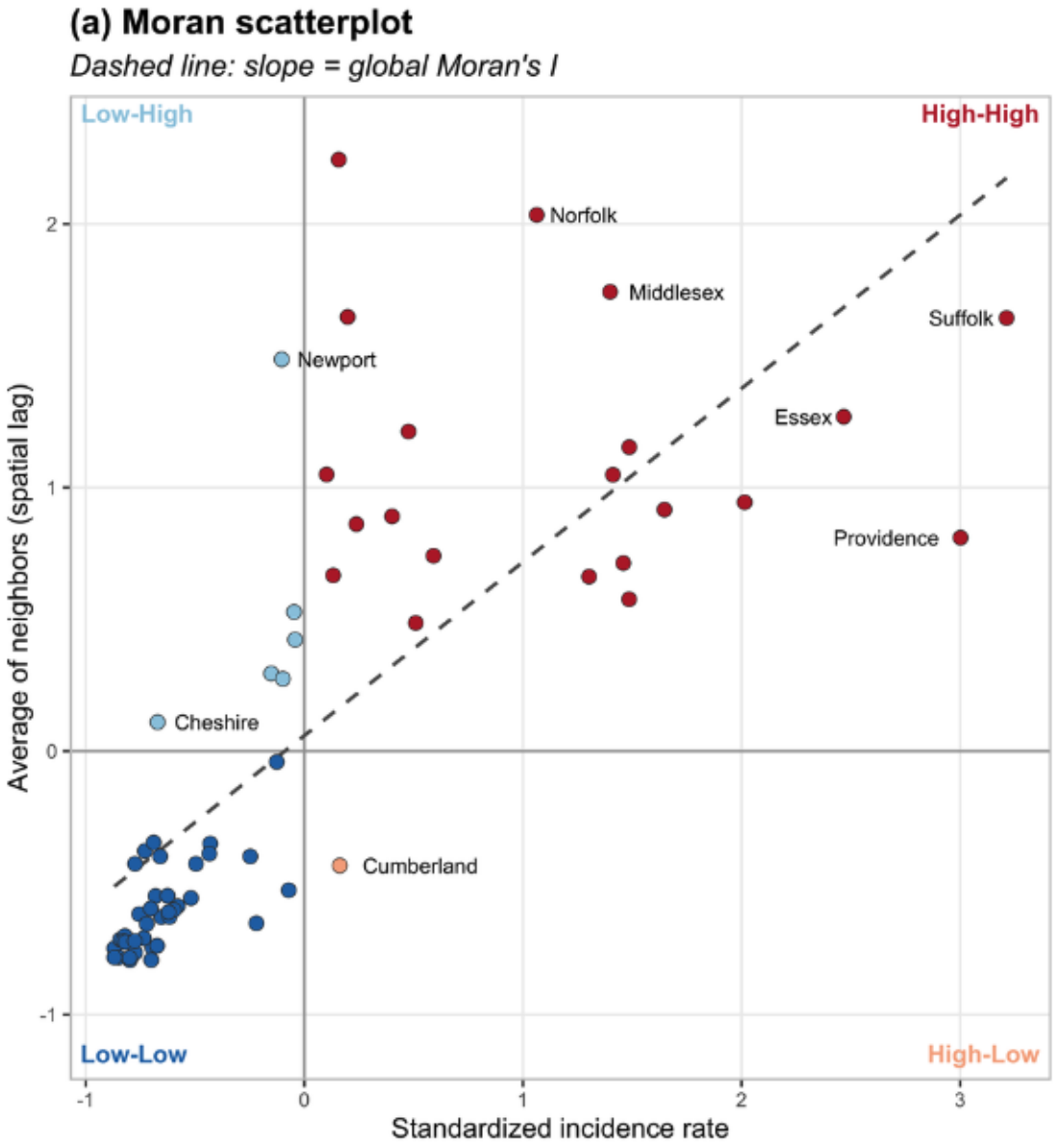
Instructions:

- Use the Moran's I equation to calculate the vibe of your neighborhood

$$I = \frac{n \sum_i^n \sum_j^n w_{ij}(y_i - \bar{y})(y_j - \bar{y})}{\left(\sum_i^n \sum_j^n w_{ij}\right) \sum_i^n (y_i - \bar{y})^2}$$

MORAN SCATTERPLOT

- x-axis: a location's standardized value.
- y-axis: the average of its neighbors (the spatial lag)
- Lines at the mean divide the plot into four quadrants
- Slope of the fitted line = global Moran's I



IMPLEMENTATION IN R

r

```r
nb <- poly2nb(pl, queen = TRUE, snap = NULL)
lw <- nb2listw(neighbours, glist = NULL, style = "W", zero.policy = NULL)
```

- **pl** — The polygons, such as an sf layer of tracts. poly2nb() returns a *neighbor list* recording which polygons border which. It holds no weights yet.
- **queen** — The contiguity rule. FALSE returns a rook-based matrix
- **snap** — A distance in map units to return borders withing a tolerance of the polygon edge
- **style** — how the nb2listw turns neightbors into weights. “B” gives every neighbor a binary weight of 1, “W” row standardizes each row so it sums to 1. Default is “W”.

IMPLEMENTATION IN R

```r
moran.test(x, listw, randomisation = TRUE, alternative = "greater")
moran.mc(x, listw, nsim, alternative = "greater")
```

- **x** — A numeric vector of attribute values, one per location, in the same order as the neighbors list. The function centers the values ($z_i = x_i - \bar{x}$).
- **listw** — the spatial weights object built with nb2listw(). The statistic is only as meaningful as this matrix, and changing it can change I.
- **randomization** —how moran.test() computes Var[I] for the z-score. TRUE (the default) assumes every rearrangement of the observed values over the polygons is equally likely. FALSE assumes the values come from a normal distribution.
- **nsim** — the number of random rearrangements moran.mc() makes, recomputing I each time.
- **alternative** - the direction of the test. "greater" (the default) tests for positive spatial autocorrelation (clustering), "less" tests for negative autocorrelation (dispersion), and "two.sided" tests for either.

Getis and Ord's Global Statistic

$$G = \frac{\sum_i^n \sum_j^n w_{ij} x_i x_j}{\sum_i^n \sum_j^n x_i \, x_j}$$

IMPLEMENTATION IN R

```r
globalG.test(x, listw, alternative = "greater")
```

- **x** — A numeric vector of attribute values, one per location, in the same order as the neighbors list. The values must be positive with a natural zero, such as counts or rates. Unlike Moran's I, G does not center the values: it multiplies them as they are.
- **listw** — The spatial weights object (built with nb2listw()). This is where the analyst's definition of "local" enters the calculation.
- **alternative** — The direction of the hypothesis test: "two.sided" (default, hot *or* cold spots), "greater" (hot spots only), or "less" (cold spots only).

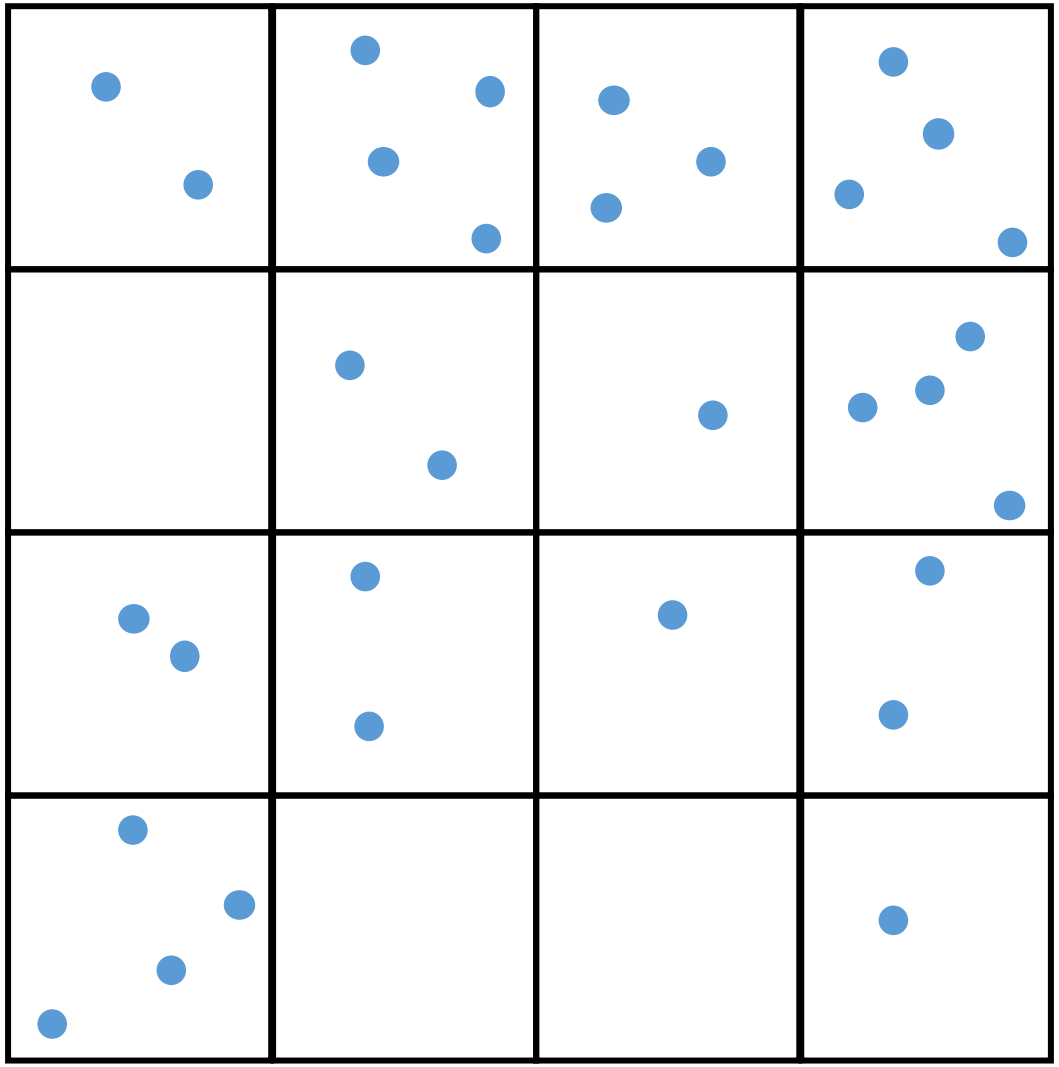
The function returns G, E[G], Var[G] and the z-score. A significant **positive z** means high values sit next to each other more than chance would produce. A significant **negative z** means they sit together less than chance would produce, usually because low values cluster.

Global Statistics

From concept to computer

Bonus Materials

Some other things to consider



QUADRAT TEST

Place a grid over the study are and assess whether the observed distribution follow the expectation under CSR

$$VMR = \frac{\sum_{=1}^{m}(x_i - \bar{x})^2}{(m-1)\bar{x}}$$

$$VMR = \frac{32}{15 \times 2} = 1.067$$

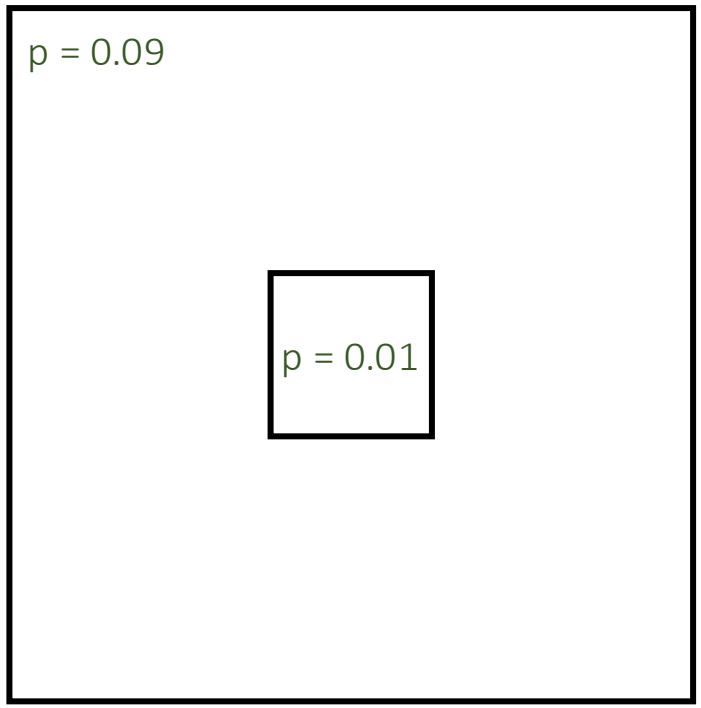
$$\chi^2_{Obs} = (16-1)1.067 = 16$$

$$\chi^2_{0.05,\, df=15} = 25.0$$

MATCHED FILTER THEOREM

An Illustrative Experiment

To illustrate the matched filter theorem, *n*=80 points were placed in a unit square with a probability of 0.9 of being randomly placed, and a probability of 0.1 of being placed in a central sub-region bounded by x- and y- coordinate 0.4 and 0.6. Grids of sizes varying from 2x2 to 10x10 were overlaid onto the study region and a quadrat test was used to assess complete spatial randomness. The procedure was repeated 10,000 times



Grid Size	Power
2 x 2	0.053
3 x 3	0.356
4 x 4	0.117
5 x 5	0.597
6 x 6	0.191
7 x 7	0.283
8 x 8	0.275
9 x 9	0.227
10 x 10	0.338